

# Limitations and Future Work

## Project: RoomRadar QC

### Purpose

This document explains the current limitations of RoomRadar QC and identifies future improvements that could make the system more accurate, automated, and useful in real campus conditions.

RoomRadar QC is currently designed as a schedule-based room suggestion system. It helps students identify rooms that do not appear to have scheduled classes during a selected day and time range. However, it does not directly measure whether a room is physically open, unlocked, or occupied in real time.

## 1. Current System Scope

The current version of RoomRadar QC allows users to search for rooms based on:

- Day
- Start time
- End time
- Building

The system checks uploaded class schedule data and returns rooms that do not have a scheduled class conflict during the selected time interval.

The current system supports:

| Feature                      | Current Status |
|------------------------------|----------------|
| Search by day                | Included       |
| Search by start and end time | Included       |

|                                                     |                    |
|-----------------------------------------------------|--------------------|
| Filter by building                                  | Included           |
| View room schedule details                          | Included           |
| Admin CSV upload                                    | Included           |
| Firebase-protected admin upload                     | Included           |
| Same-day room search                                | Included           |
| Live occupancy tracking                             | Not included       |
| Room reservations                                   | Not included       |
| User room access reports                            | Not included       |
| Room confidence score                               | Not included       |
| Automated schedule crawler/importer                 | Not included       |
| Verified lab-room exclusion list                    | Not included       |
| Date-specific search for holidays/finals exceptions | Future improvement |

## 2. Important Clarification: Schedule-Based Suggestions

RoomRadar QC does not guarantee that a room is physically available.

The system determines whether a room has a scheduled class conflict. If no class is scheduled, the room may appear as a suggested room. However, no scheduled class does not always mean the room is actually available for student use.

A room may still be unavailable because:

- The room is locked.
- The room is occupied by students, faculty, staff, or another group.
- The room is reserved for a club, event, meeting, or department use.
- The room is restricted, such as a lab or specialized classroom.
- The room is closed because of a holiday, maintenance, or campus schedule change.
- The uploaded schedule data is outdated or incomplete.

For this reason, RoomRadar QC results should be understood as **schedule-based room suggestions**, not guaranteed real-time availability.

The current wording in the project is intended to communicate that RoomRadar QC helps users identify rooms that appear open according to schedule data. It does not claim to physically confirm whether a door is unlocked or whether people are inside the room.

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## 3. Limitation: No Direct Real-Time Occupancy Detection

### Current Limitation

The system does not currently know whether people are physically inside a room.

RoomRadar QC can check whether a class is scheduled, but it cannot directly detect whether a room is occupied at that exact moment.

There is no direct software-only way for the system to know whether someone is inside a classroom unless the campus installs some kind of physical detection method or provides official real-time room access data.

Examples of direct occupancy methods would include:

- Motion sensors
- Occupancy sensors
- Radar-based sensors
- Door access sensors
- Camera-based systems
- Official room check-in hardware
- Official live building access systems

Installing sensors in every classroom would require campus-level approval, hardware installation, privacy review, maintenance, and cost. That is outside the scope of the current class project.

## Future Improvement

A future version could improve real-world accuracy through indirect methods, such as user self-reporting. Instead of claiming direct occupancy detection, users could report whether a suggested room was actually usable.

Possible report options:

- Room was open and usable.
- Room was locked.
- Room was occupied.
- Room was restricted.
- Room was being used for an event.
- Schedule information seemed incorrect.

This would not be as exact as physical sensors, but it could help the system become more useful over time by collecting real student feedback.

## 4. Limitation: Lab and Restricted Room Identification

### Current Limitation

The system may return rooms that do not have scheduled classes but are not realistic study spaces. For example, a lab room may appear free based on schedule data, but it may be locked or restricted unless an instructor or authorized staff member is present.

This is difficult to fully solve because the team does not currently have a direct official list of all lab rooms, restricted rooms, or department-controlled rooms on the Queens College campus. Without that verified list, the system cannot reliably exclude every lab or restricted room from search results.

The current system only has the room and schedule information available from the uploaded

semester data. That data can show where classes take place, but it does not always clearly say whether a room is a general classroom, science lab, computer lab, department-only room, or restricted space.

Because of this, automatically excluding all lab rooms is not reliable without additional official room metadata.

## Future Improvement

A future version could improve this by adding verified room metadata.

Possible sources of room metadata could include:

- An official campus facilities room inventory
- Department-provided room access lists
- Manually verified room categories
- Admin-maintained room labels
- User-reported room access information

Possible room categories:

- General classroom
- Lecture hall
- Computer lab
- Science lab
- Department-restricted room
- Study-friendly room
- Unknown access room

Until a verified room-type list is available, the safer approach is to clearly label results as schedule-based suggestions instead of claiming guaranteed availability.

Example future label:

Science Building A135B — No class scheduled, but room access may be restricted.

This would help users understand that the room has no scheduled class conflict but may still not be a practical study space.

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## 5. Limitation: Manual CSV Upload

### Current Limitation

The current system depends on an admin uploading semester schedule data through the admin dashboard.

This approach works for the current version because it gives the admin control over which semester data is uploaded. However, it also creates limitations:

- The admin must remember to upload new semester data.
- The admin must make sure they are uploading the most recent semester data.
- If the course schedule changes after upload, the system may not immediately reflect those changes.
- The schedule database depends on the quality and freshness of the uploaded CSV file.

### Future Improvement: Automated Import Pipeline

A future version could automate the schedule update process.

Possible future architecture:

Scheduled Task



Web Crawler or Data Fetcher



CSV / Schedule Data Retrieval



Parser



Database Update



Room Search Uses Updated Data

This would reduce manual work and help keep the database closer to the latest available

schedule data.

Potential benefits:

- Less manual admin effort
- More current schedule data
- Reduced risk of outdated uploads
- Better long-term maintainability

Potential challenges:

- The source data format may change.
- The crawler may break if the school website changes.
- The system would need error handling and logging.
- The import process would need to avoid duplicate records.
- The system would need rules for updating or replacing old semester data.
- The system would need to confirm that automated access to the data source is allowed.

For the current version, manual CSV upload was used because it was simpler to validate and control during the project timeline. Automated importing is a strong future improvement, but it would require additional reliability and maintenance planning.

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## 6. Design Decision: Same-Day Search Instead of Multi-Day Date Range Search

### Current Design

RoomRadar QC uses a same-day search model. Users select:

- Day
- Start time
- End time
- Building

This design is intentional because the app is meant to help students find a room for a realistic study period during the same campus day.

For example:

Monday, 12:30 PM to 3:30 PM

The system assumes that the selected start time and end time are on the same day.

## Rationale

A multi-day room search, such as:

May 25 at 9:00 AM to May 27 at 2:00 PM

does not match the normal use case for this system.

RoomRadar QC is not meant for overnight room use, long-term room occupation, or multi-day reservations. Students are not expected to stay in a classroom across multiple days, and campus buildings close at night. Because of this, an end date would add complexity without supporting the main student workflow identified in the interviews.

The intended use case is:

I am on campus today, or I know I will be on campus on a specific weekday, and I want to find a room during a specific time interval.

For that reason, the current same-day search model is appropriate for the projects scope.

## Possible Future Improvement

A future version could add date selection, but mainly for date-specific accuracy rather than multi-day room use.

For example, date selection could help account for:

- Holidays



- Special campus closures
- Finals week schedule changes
- Semester start and end dates
- Special event days

In that future version, the user could select a specific date, but the search would still normally remain within a same-day time interval.

Example future search:

May 25, 12:30 PM to 3:30 PM

This would be more useful than a multi-day search because it supports real campus behavior while still keeping the interface simple.

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## 7. Clarification: Room Details Schedule Is Optional Context

### Current Design

When a user searches for a time slot, RoomRadar QC returns a list of room cards that appear available based on the selected search parameters.

Each room card represents a room that does not have a scheduled class conflict during the users selected day and time interval.

The **View Room Schedule** button is optional. It allows the user to see more context about that rooms schedule for the rest of the day.

### Why the Schedule Details Are Shown

The Room Details page is not meant to contradict the search result. It is meant to provide additional information if the user wants to check the rooms schedule before going there.

For example, a user might search:

Monday, 12:30 PM to 3:30 PM

If a room appears in the results, that means the room has no scheduled class conflict during that selected interval.

If the user clicks **View Room Schedule**, the app may show classes before or after the searched time. Those schedule entries are not saying the room is unavailable during the selected search interval. They are simply showing the rooms broader schedule for context.

This supports one of the needs found during user interviews: students wanted to know how long they could realistically stay in a room before another class might begin.

## UI Improvement

Although the current design is intentional, the wording can be improved so users better understand the difference between:

- The users selected search interval
- The rooms full schedule for the day
- The optional schedule context shown on the Room Details page

Possible wording for search results:

Suggested for your selected time: 12:30 PM to 3:30 PM

Possible wording for the Room Details page:

Full room schedule for Monday

Possible helper text:

This schedule is shown for context. The room appeared in your search because it had no scheduled class conflict during your selected time interval.

This would make the UI clearer without changing the main design.

## 8. Future Feature: User Self-Reporting

Some interview responses suggested that students want more than schedule data. Students mentioned ideas such as live counters, check-ins, room status updates, and knowing whether rooms are actually occupied.

Because the system cannot directly know if someone is in a room without physical sensors or official room access data, user self-reporting would be a practical software-based improvement.

Users could report:

| Report Type         | Meaning                                                   |
|---------------------|-----------------------------------------------------------|
| Open and usable     | The room was available and could be used.                 |
| Locked              | The room was physically locked.                           |
| Occupied            | People were already inside.                               |
| Restricted          | The room appeared to be a lab or restricted space.        |
| Event in progress   | The room was being used for an event or meeting.          |
| Inaccurate schedule | The room had a class or activity not shown by the system. |

This would help improve trust in the results over time.

A self-reporting feature would not be perfect because it depends on users submitting accurate reports. However, it would still provide more real-world feedback than schedule data alone.

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## 9. Future Feature: Confidence Score

A future version could calculate a confidence score for each room suggestion.

The confidence score would not claim to know true occupancy directly. Instead, it would estimate the trustworthiness of a room suggestion based on schedule data and user feedback.

Example:

Room: KY 250

Schedule Status: No class scheduled

Confidence: High

Reason: Users recently reported successful access.

Another example:

Room: SB A135B

Schedule Status: No class scheduled

Confidence: Low

Reason: Frequently reported as locked or restricted.

Possible confidence factors:

- Whether a class is scheduled
- Whether the room has been reported open recently
- Whether the room is often reported locked
- Whether the room is a lab or restricted space
- Whether the room is frequently occupied outside class time
- Time of day
- Day of week
- Recent user reports

This would directly address the issue that no scheduled class does not always mean the room is truly available.

The confidence score would be an estimate, not a guarantee. It would make the search results more transparent by showing users how reliable a suggestion appears to be.

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## 10. Future Feature: Room Type and Room Details

Some students wanted more information about the room before choosing it, such as whether it is a large lecture hall, small classroom, or has certain features.

Future room details could include:

- Room type
- Room capacity
- Pictures
- Desk/chair layout
- Lab or non-lab status
- Whiteboard/projector availability
- Accessibility information
- Whether the room is usually open to students

This would make search results more useful and help users avoid rooms that technically have no class scheduled but are not good study spaces.

This improvement depends on having a reliable source of room metadata. Since there is no simple direct list of all lab or restricted rooms currently available to the team, this would likely require manual verification or official campus data.

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## 11. Future Feature: Notifications

Some interview feedback suggested notifications as a useful feature. A future version could allow users to be notified when a room becomes available or when a selected time is approaching.

Possible notification ideas:

- Notify when a room becomes free based on schedule data.
- Notify before a scheduled class starts in the room.
- Notify when a reported room status changes.
- Notify when a preferred building has suggested rooms.

This feature would require user accounts or notification permissions, so it was not included in the current version.

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## 12. Future Feature: Reservation Support

Some students mentioned that reserving a room would be useful. A reservation system would be more complex than schedule-based search because it would need official campus policy support.

A reservation feature would require:

- User accounts
- Reservation rules
- Conflict detection
- Admin approval or automatic approval
- Limits on reservation length
- Handling cancellations
- Preventing abuse
- Coordination with campus departments

Because of this, reservation support is treated as a future feature rather than part of the current project version.

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## 13. Current Strengths of the System

Although the current version has limitations, it still addresses the core user problem found in interviews.

Current strengths:

- Provides a dedicated interface instead of requiring students to search through raw schedule data.
- Allows users to search by day and time.
- Allows building-based filtering.
- Shows room suggestions based on schedule conflicts.
- Shows room schedule details as optional context.

- Provides admin upload for semester data.
- Protects admin upload using Firebase authentication.
- Separates frontend, backend, and database responsibilities.
- Uses environment variables for deployment configuration.
- Clearly identifies that results are schedule-based suggestions rather than guaranteed real-time availability.
- Keeps the search workflow simple by using a same-day search model that matches normal campus usage.

The current version should be viewed as an MVP: a first working version focused on schedule-based room discovery.

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## 14. Summary

RoomRadar QC currently provides schedule-based room suggestions. It helps students identify rooms that do not have scheduled class conflicts during a selected day and time.

The main limitation is that the system does not directly know whether a room is physically open, unlocked, unrestricted, or occupied in real time. Direct real-time occupancy detection would require physical infrastructure, such as sensors or official access-control data, which is outside the scope of the current software project.

Another limitation is that the team does not currently have a verified official list of all lab rooms or restricted rooms on campus. Because of this, the system cannot perfectly exclude every restricted room from search results.

The same-day search model is an intentional design decision. It matches the main use case of students looking for a room during a realistic same-day campus time interval. Multi-day room use does not match the normal purpose of the application, but date-specific search could be considered in the future for holidays, finals week, and campus closure exceptions.

The Room Details page is also intentional. It provides optional schedule context for users who want to understand what happens before or after their selected time slot.

Future improvements should focus on:

- Automated schedule importing
- Date-specific search for holidays, finals week, and campus closure exceptions
- Verified room type metadata
- Restricted room labeling
- User self-reporting
- Room confidence scores
- Notifications
- Reservation support